

# Design of a 2.3GHz - 2.6GHz Mixer First Receiver using 90 nm CMOS Technology

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*Academic Integrity: The following design and summary report represent my own original work. I acknowledge that actual designs must be unique. I acknowledge that absolutely no sharing of schematics, subcircuits, designs, plots, write-ups, or reports will be accepted. I have not made any use of prior-semester reports or prior-semester designs. I acknowledge that students in violation of this policy will be reported to the NC State Office of Student Conduct, with repercussions including placement on academic probation and grade penalties including but not limited to receiving zero credit for the design project.*

## Abstract:

This report presents the design process of a mixer-first receiver operating in the frequency range of 2.3 GHz to 2.6 GHz. The receiver was biased with a 2.5V DC source and 100 uA of current. There are two main components in this design, namely the four-phase passive mixer and a Transimpedance Amplifier (TIA). The baseband frequency is 50 MHz. The TIA performs gain boosting to compensate for the attenuation by the mixer. The design performed well on the linearity parameter of  $iP1_{dB}$ , obtaining a good input matching with the  $S_{11}$ , all while maintaining a noise figure below the specified metrics. However, as the attenuation of the signal from the mixer's side had a large contribution to the overall gain degradation, the TIA was not able to provide enough gain to match the conversion gain requirement, while still having a good standalone performance, leading to a negligible power consumption. However, due to a lack of other elements such as LNA and LO-buffers, certain aspects of the receiver could not be reached to their full potential. Design is from Mixer Lecture 19.

## Specification Table:

| Metric            | Specification                        | Our Design |            |            |
|-------------------|--------------------------------------|------------|------------|------------|
|                   |                                      | 2.3 GHz    | 2.45 GHz   | 2.6 GHz    |
| Frequency Range   | 2.3-2.6 GHz<br>Baseband BW<br>50 MHz | N/A        | N/A        | N/A        |
| Noise Figure(dB)  | <14dB                                | 13.46 dB   | 13.46 dB   | 13.47 dB   |
| Power Consumption | <25mW for 1V<br>and/or 2.5 V         | 675.67 uW  | 675.67 uW  | 675.67 uW  |
| $S_{11}$          | < -10dB                              | -18.16 dB  | -18.08 dB  | -17.99 dB  |
| $iP1_{dB}$        | > -15dBm                             | -8.135 dBm | -8.143 dBm | -8.145 dBm |
| Conversion Gain   | > 10dB                               | 2.22 dB    | 2.25 dB    | 2.27 dB    |

## Design Contributions:

Ali worked on designing the mixer and the receiver testbench along with optimizing the mixer for adequate performance. Abhishek was responsible for designing the trans-impedance amplifier and optimizing it to help boost the performance of the mixer to satisfactory levels.

## Design Overview:

### Mixer:

The mixer design is a single-balanced 4-phase passive mixer with a 25% duty cycle. The passive mixer has a very simple design and is able to draw in less DC power than other mixer topologies, which provide plenty of room for power consumption. A single-balanced mixer has benefits that include high RF injection at the IF port, rejection of mixing spurs, and suppression of LO noise. Unfortunately, the passive mixer attenuates signals that make it difficult to achieve very high conversion gain, requiring very high LO drive requirements. A 4-phase mixer was the best choice for our mixer topology as it avoids image current flowing through the receiver. Our topology consisted of 4 1V NMOS transistor switches and 4 baseband capacitors. The RF input signal is converted into an IF signal using switching transistors, which can cut the RF current when the LO frequency is set.

### TIA:

For the TIA, the topology selected was an active-load differential amplifier with a single-ended output. Initial dimensions at a biasing point were allocated to the devices and the TIA was put into the main receiver's hierarchy to test out whether all devices are in saturation or not. Using these tests, an optimum biasing point and the corresponding dimensions. It was observed that increasing the number of fingers greatly improved the noise, but degraded the gain, hence, a valid tradeoff between these had to be found. An alternative topology with the active load was also tried, but it failed to get all devices in saturation, thus failing the noise performance. Furthermore, introducing a feedback resistor also did not cause drastic improvements.

## Design Methodology:

When creating the mixer, we first wanted to check that our topology was mixing at different frequencies. Once it was capable of performing its basic operation, we moved onto sizing the switches within the passive mixer. With sizing, it was important that we keep the switch resistance less than 5 ohms because having high switch resistance results in a high noise figure. To adjust for optimal switch resistance you will need to increase the size of the fets in the mixer and configure the RC networks connected to the output of the mixer. Switch resistance is found by setting the baseband resistance low and shunt resistance is found by setting it high. By using the best combination of fet size and RC network values, you can not only have complete input impedance matching but also low noise figure. The tradeoff for our design is that when the input impedance matched, the noise figure was too high. The tradeoff had to be made to sacrifice some input impedance matching to decrease the noise figure to an acceptable value by finding the optimal fet size. In the TIA, the primary objective was to find a suitable operating point, for which the Vgate was swept, while alternative phased inputs were provided to the NMOS. The goal was to source maximum current from the active load and get a sufficient  $r_o$  to keep the devices saturated, while still getting a decent gain and noise. To do this, eventually, the number of fingers were set to 1, the length was constant for all devices, and the total width was varied, thus providing a good standalone gain.

( $A_v = g_{m1}(r_{o2} || r_{o4})$ ). Thus minimally sizing the length while increasing the total width was a viable working solution.

## Schematics:

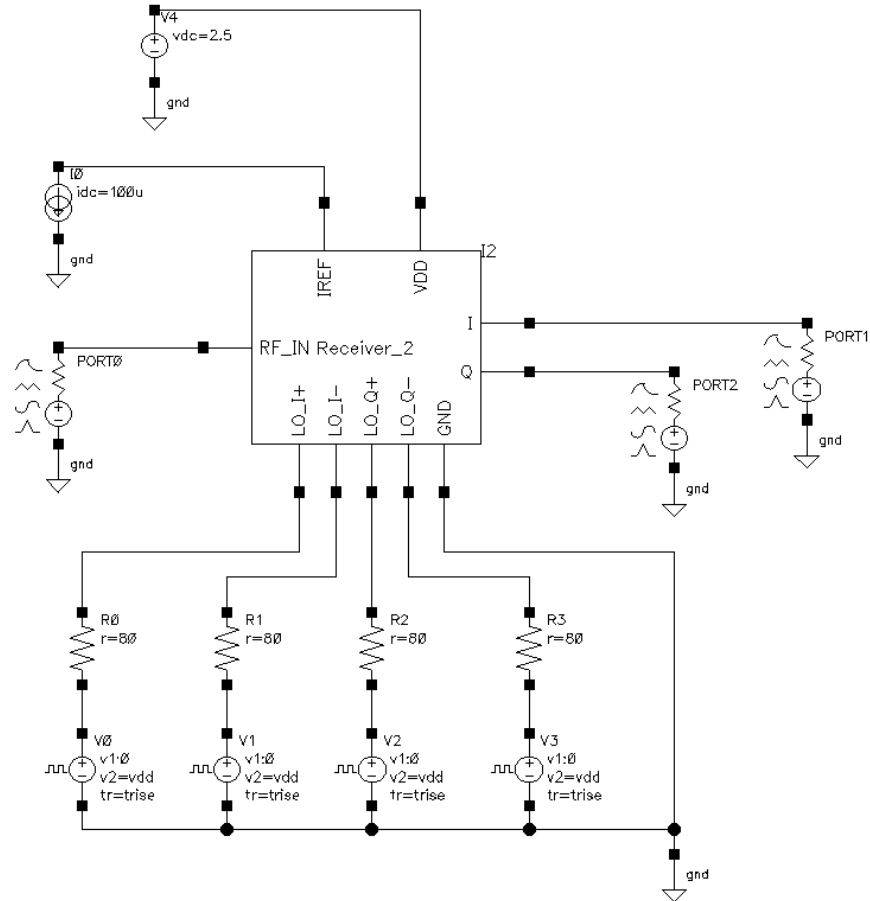


Figure 1: Receiver Testbench



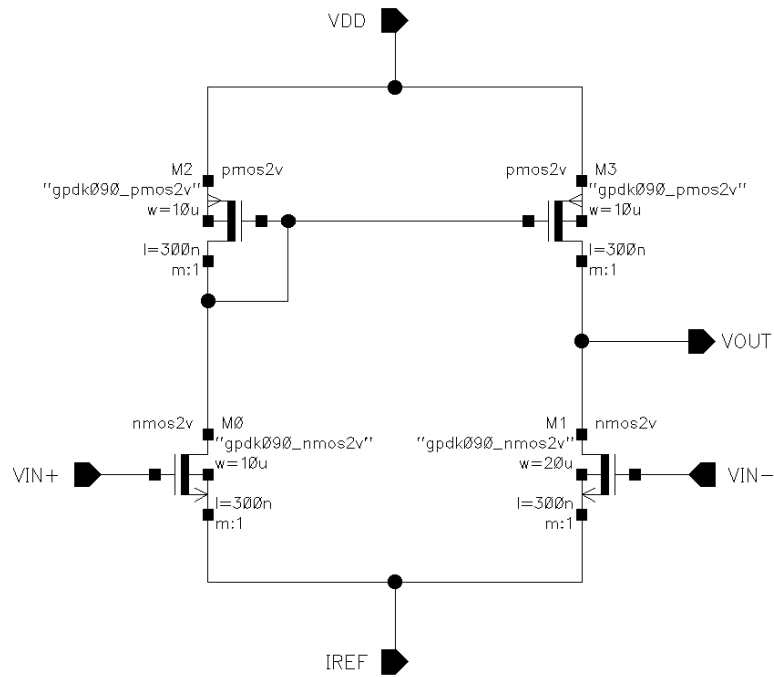


Figure 4: TIA Schematic

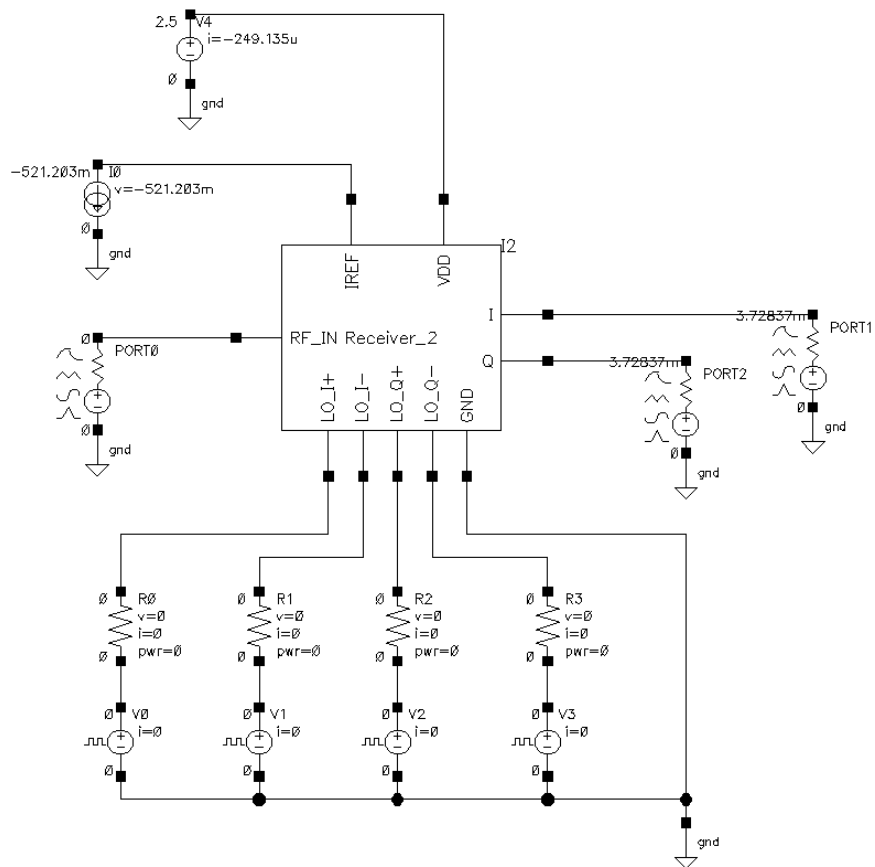


Figure 5: Receiver Testbench w/DC Operating Points and Node Voltages

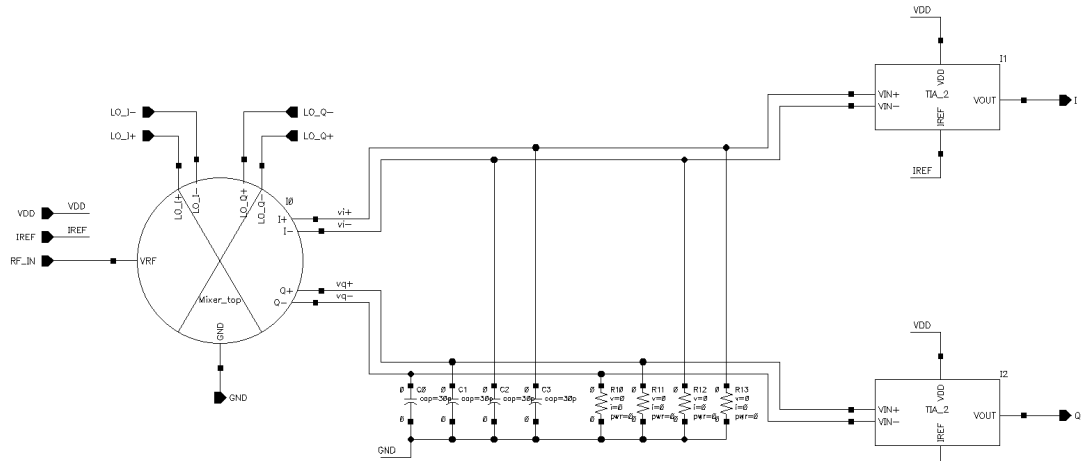


Figure 6: Receiver Schematic w/DC Operating Points and Node Voltages

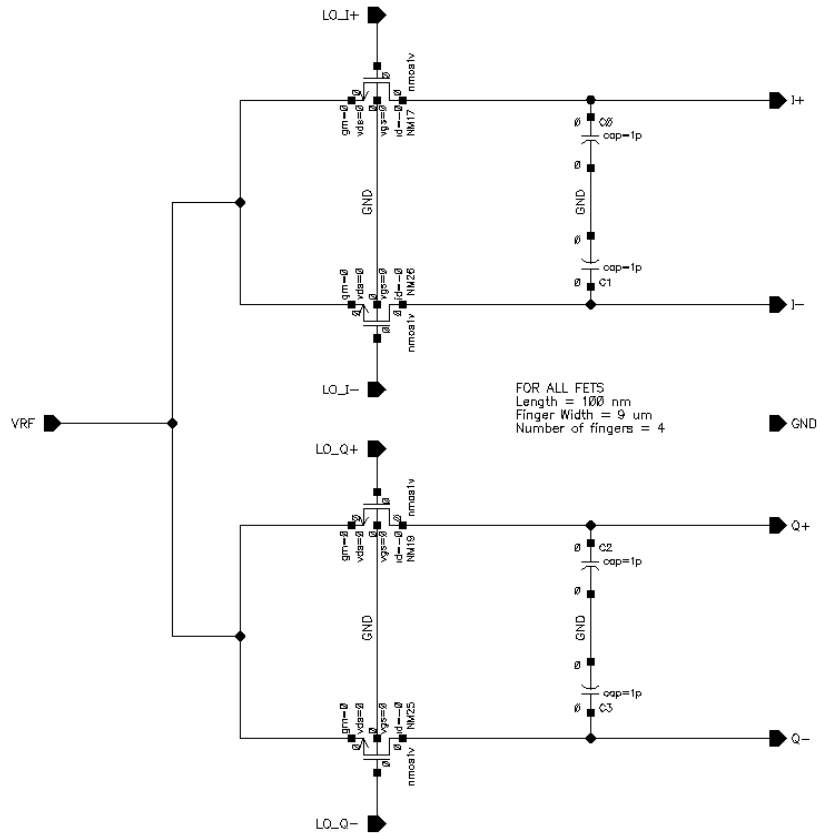


Figure 7: Mixer Schematic w/DC Operating Points and Node Voltages

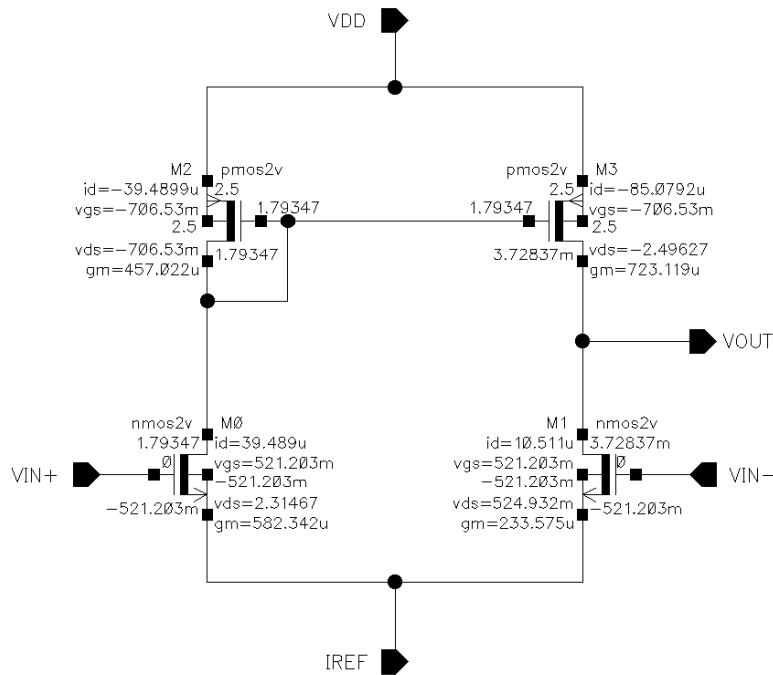


Figure 8: TIA Schematic w/DC Operating Points and Node Voltages

## Performance Plots for the Mixer:

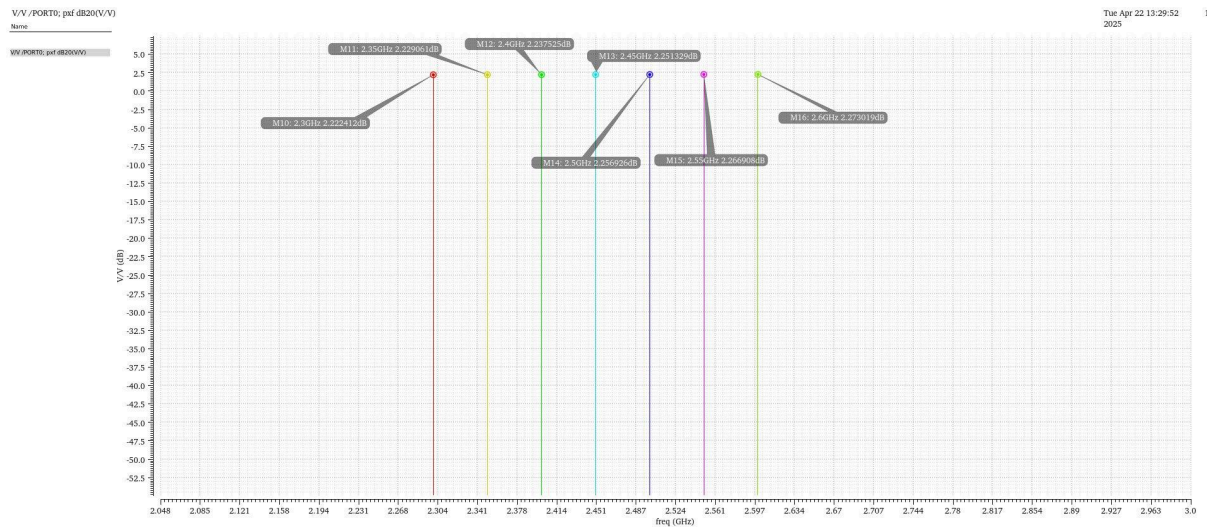


Figure 9: Conversion Gain for Swept LO and Fixed IF at 2.3 GHz

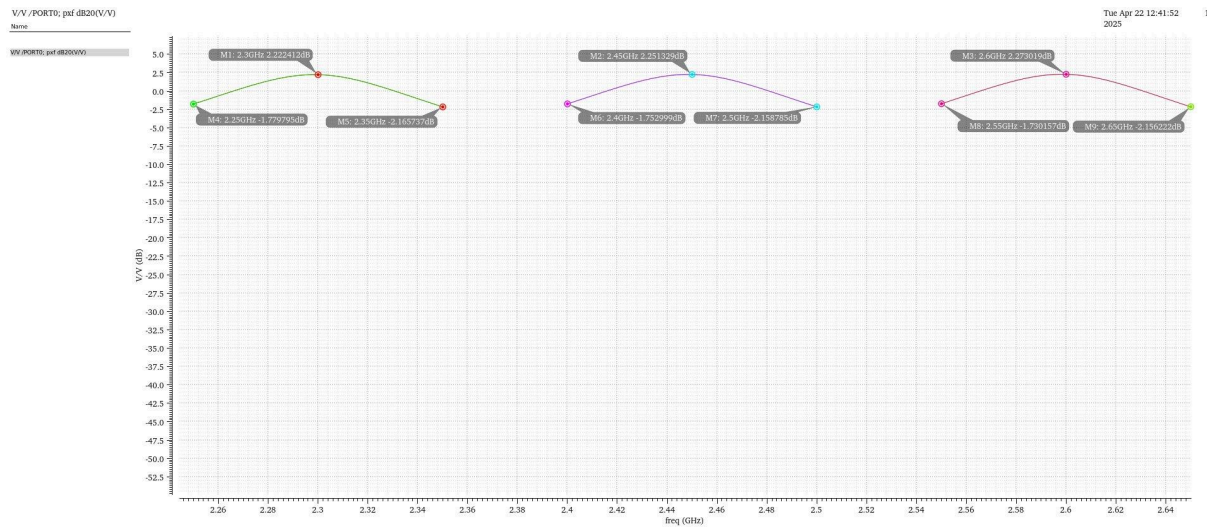


Figure 10: Conversion Gain for Swept IF and Fixed LO at 2.3 GHz

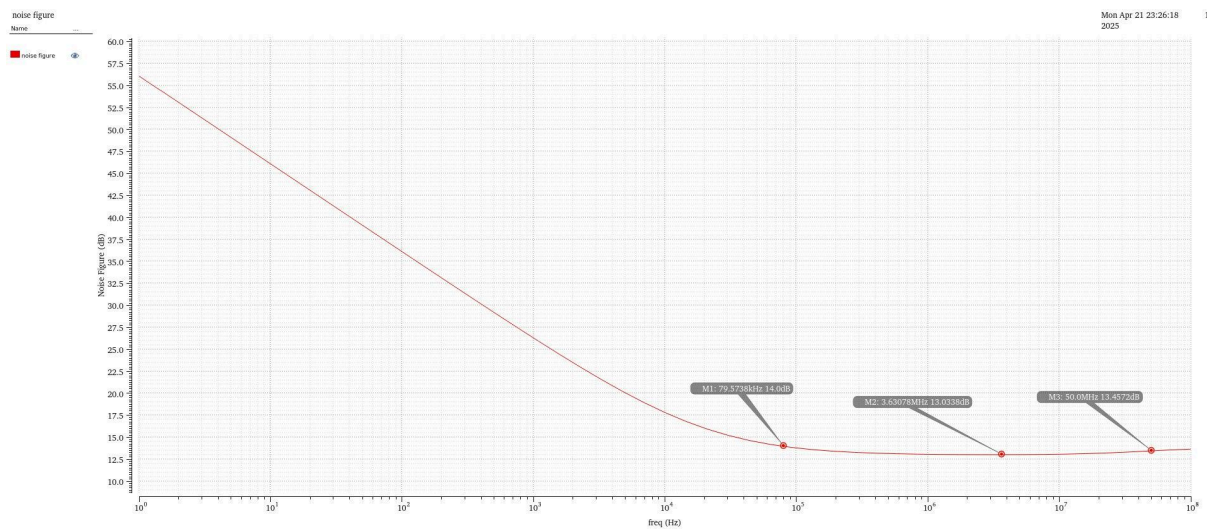


Figure 11: Noise Figure vs Baseband Frequency at 2.3 GHz (Meets requirement at baseband frequency. Flicker noise occurs between 0-79 kHz)

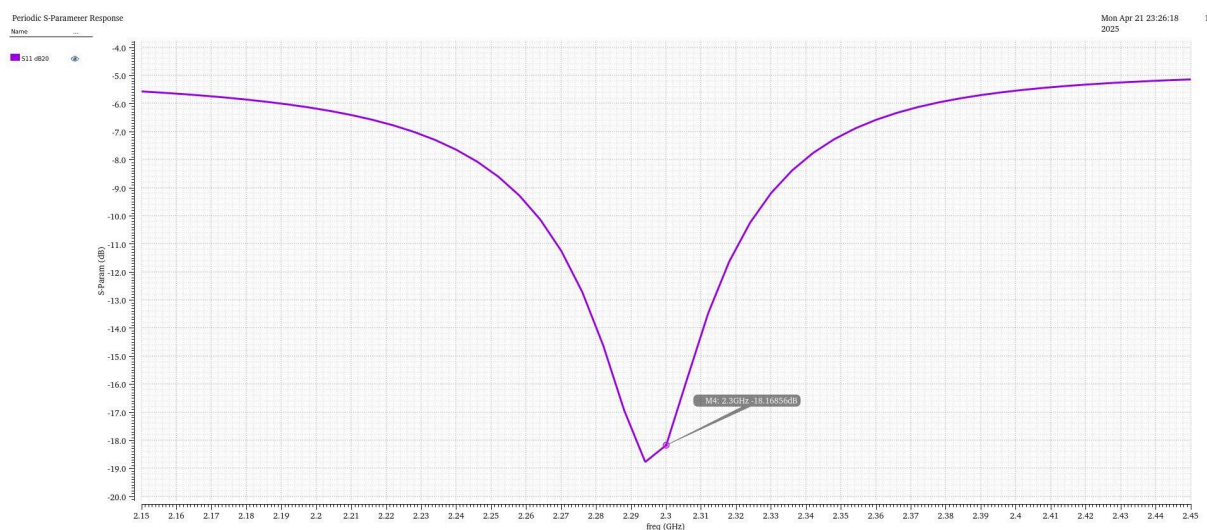


Figure 12: S11 Magnitude vs Frequency at 2.3 GHz

S11  
Name: V1r  
S11

Mon Apr 21 23:50:58  
2023 1

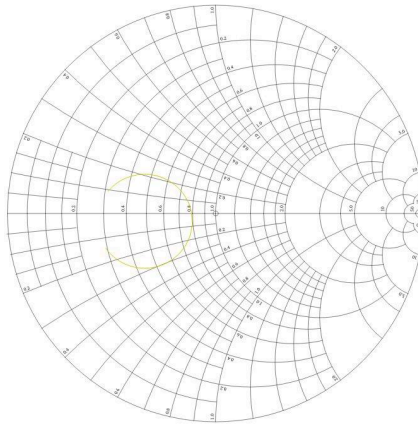


Figure 13: S11 Smith Chart at 2.3 GHz

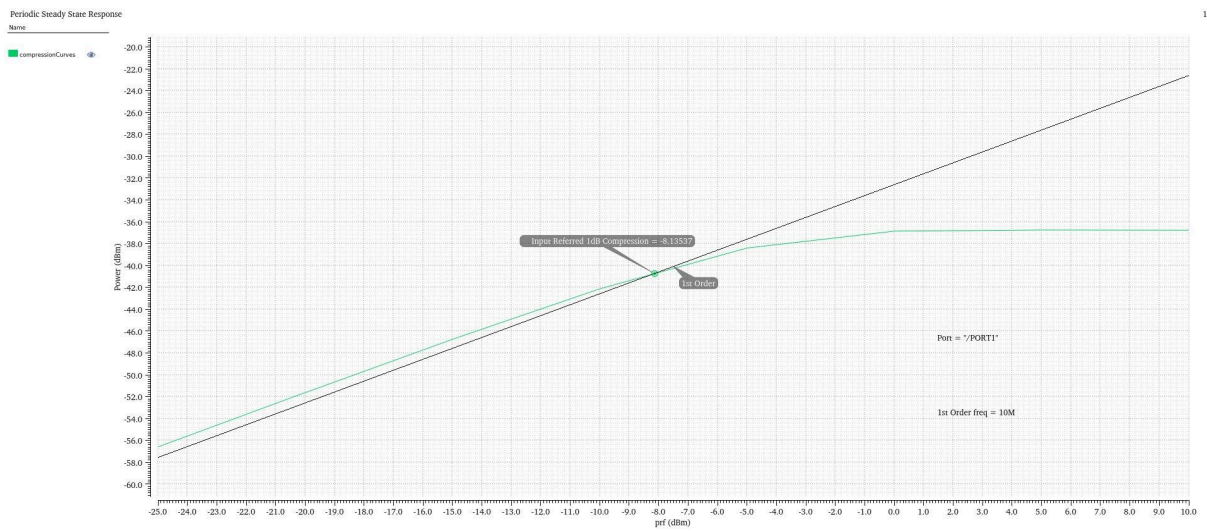


Figure 14: Input Referred P1dB Compression Point at 2.3 GHz

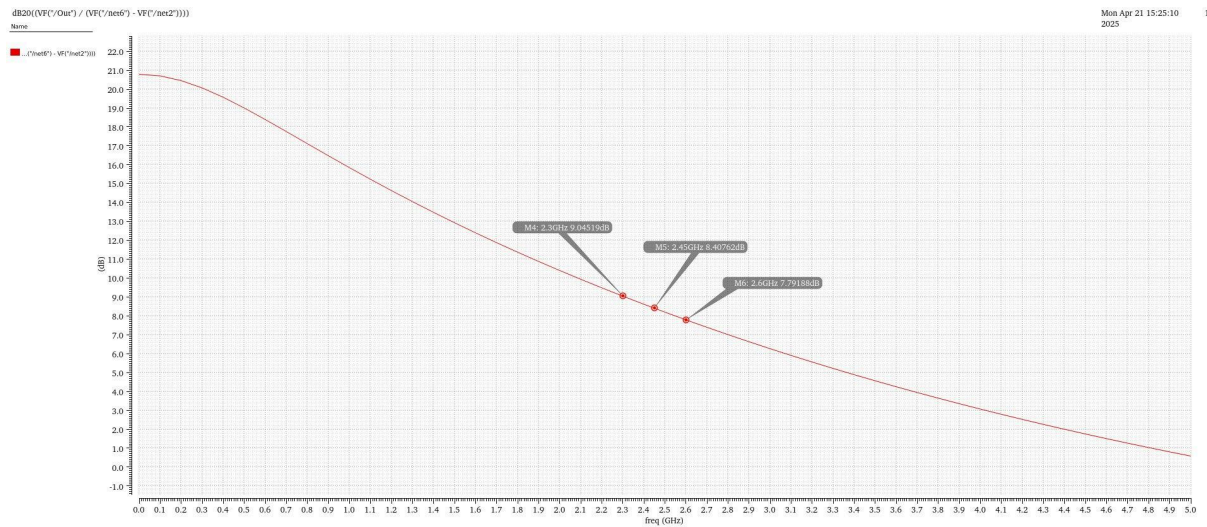


Figure 15: AC Gain for Standalone Baseband Amplifier at 2.3 GHz

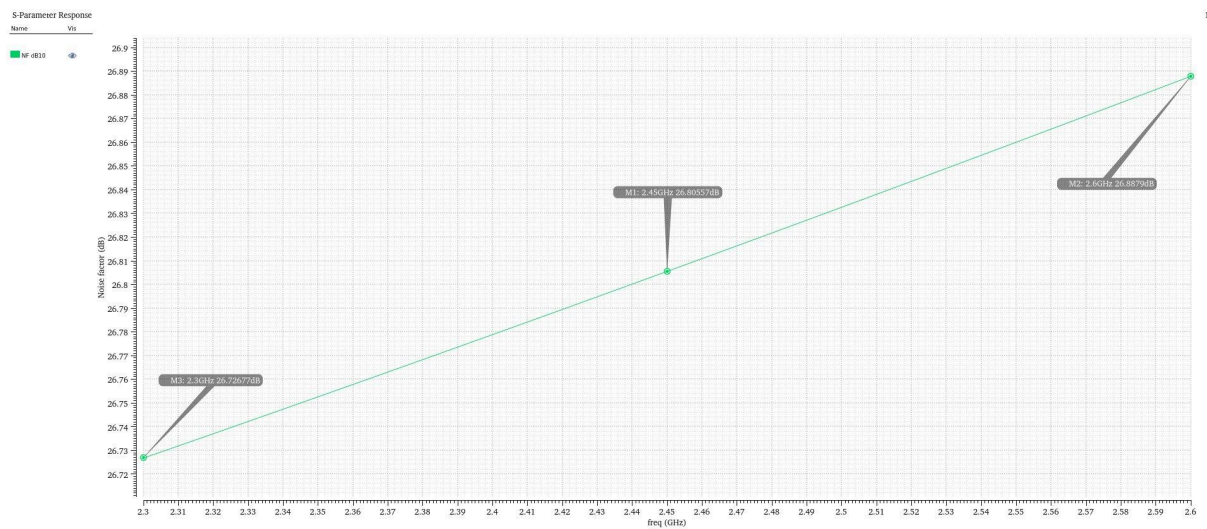


Figure 16: Noise for Standalone Baseband Amplifier 2.3 GHz

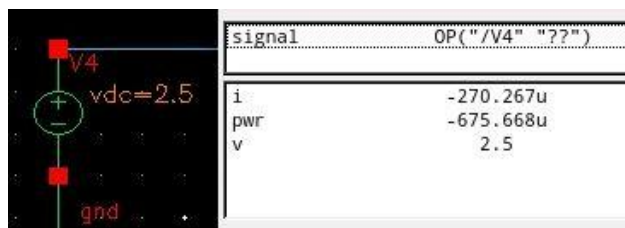


Figure 17: Power Consumption at 2.3 GHz

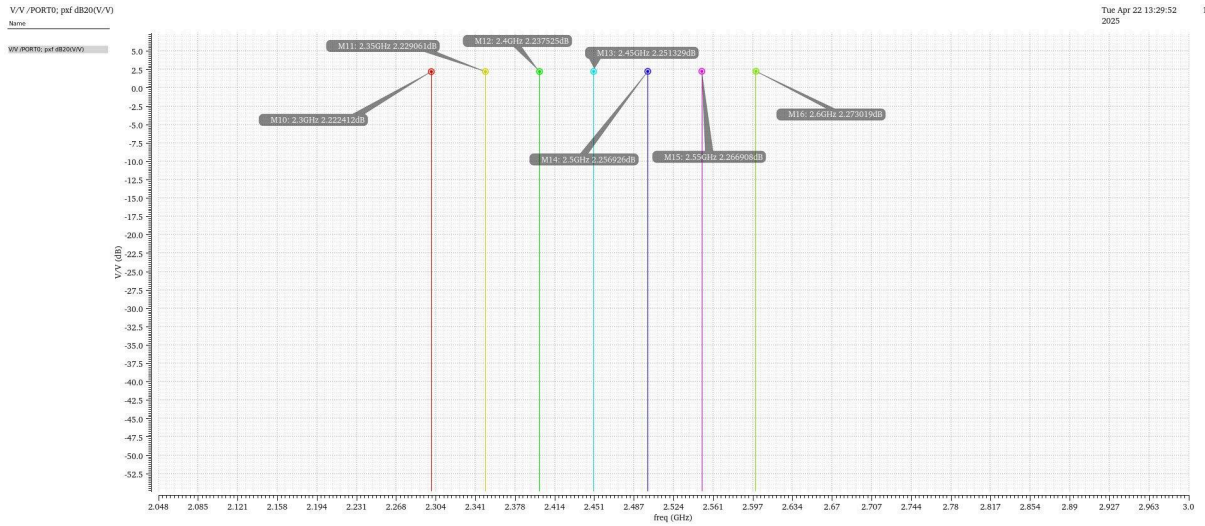


Figure 18: Conversion Gain for Swept LO and Fixed IF at 2.45 GHz

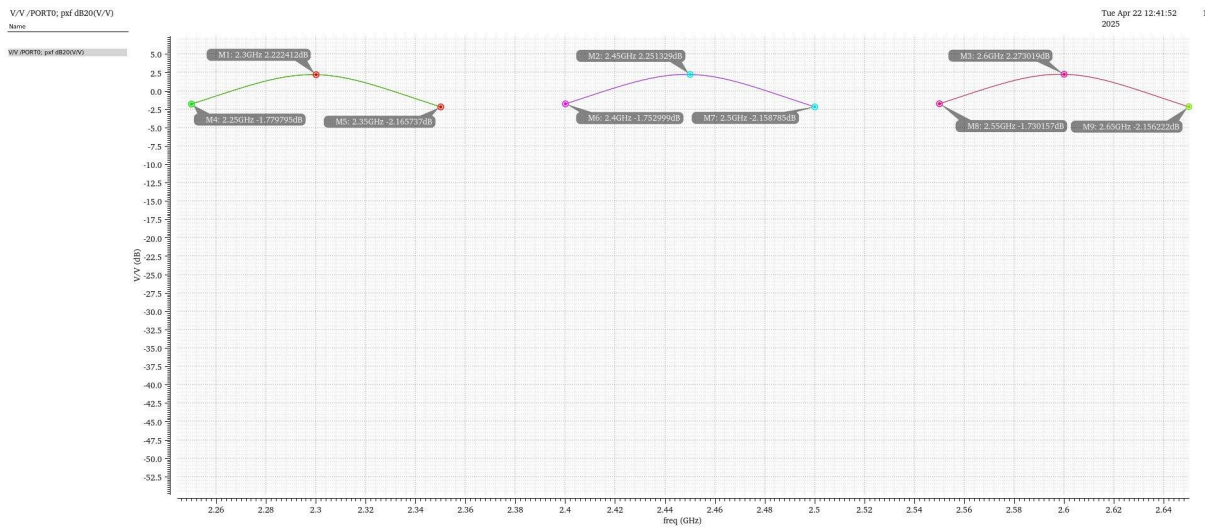


Figure 19: Conversion Gain for Swept IF and Fixed LO at 2.45 GHz

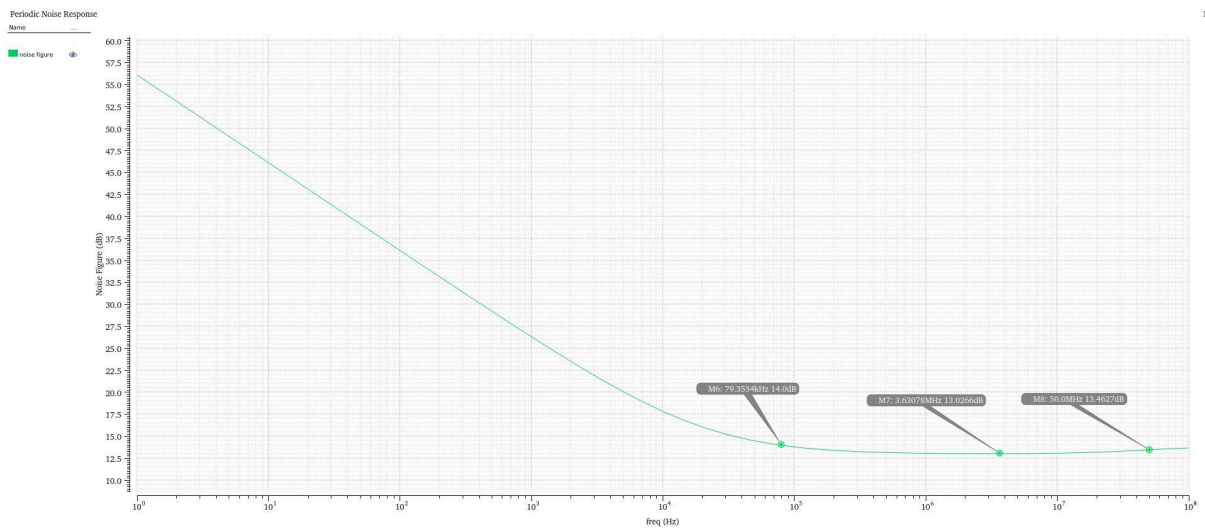


Figure 20: Noise Figure vs Baseband Frequency at 2.45 GHz (Meets requirement at baseband frequency. Flicker noise occurs between 0-79 kHz)

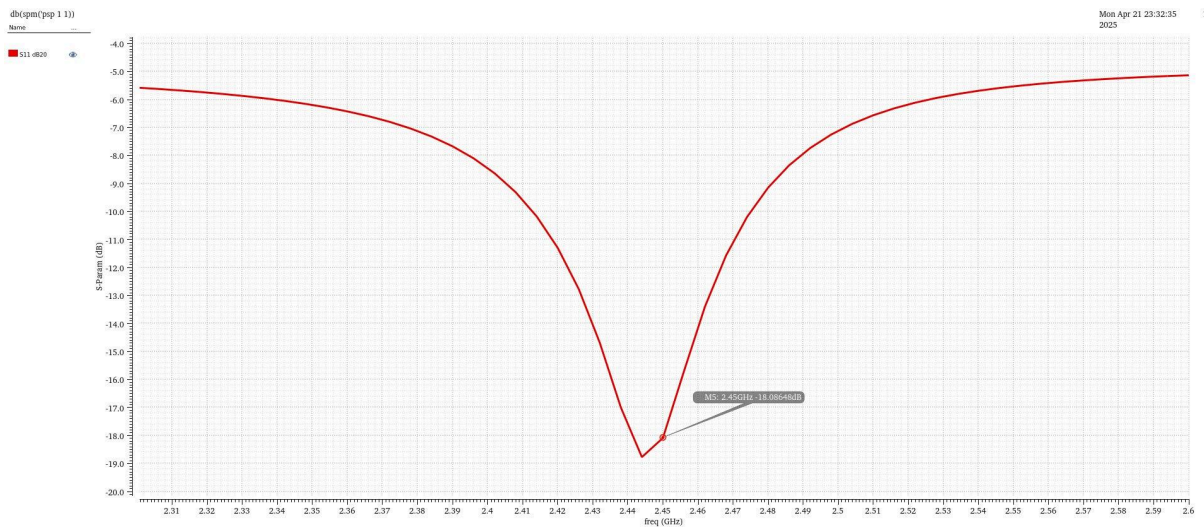


Figure 21: S11 Magnitude vs Frequency at 2.45 GHz

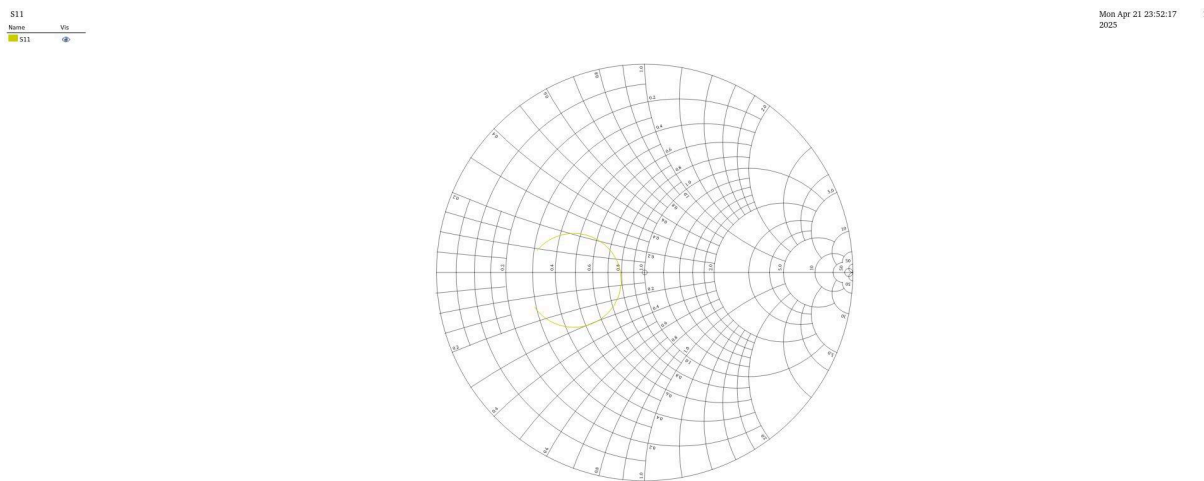


Figure 22: S11 Smith Chart at 2.45 GHz

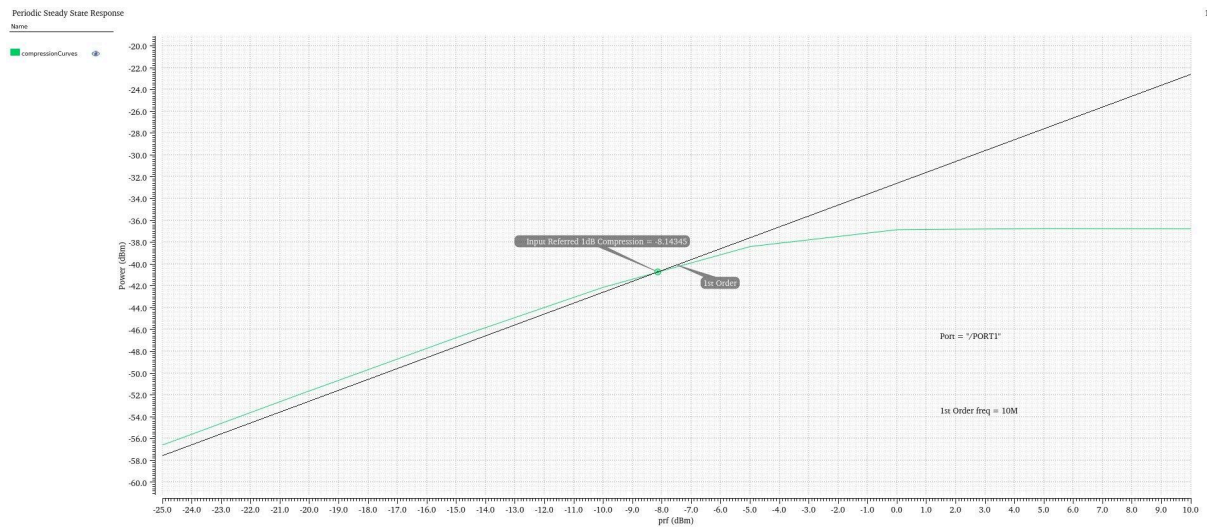


Figure 23: Input Referred P1dB Compression Point at 2.45 GHz

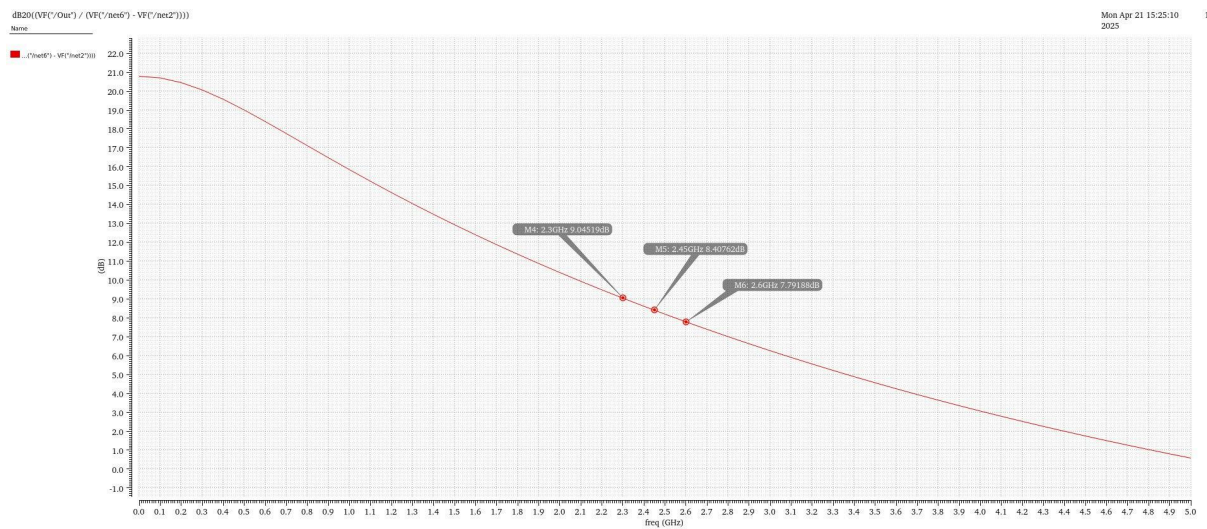


Figure 24: AC Gain for Standalone Baseband Amplifier at 2.45 GHz

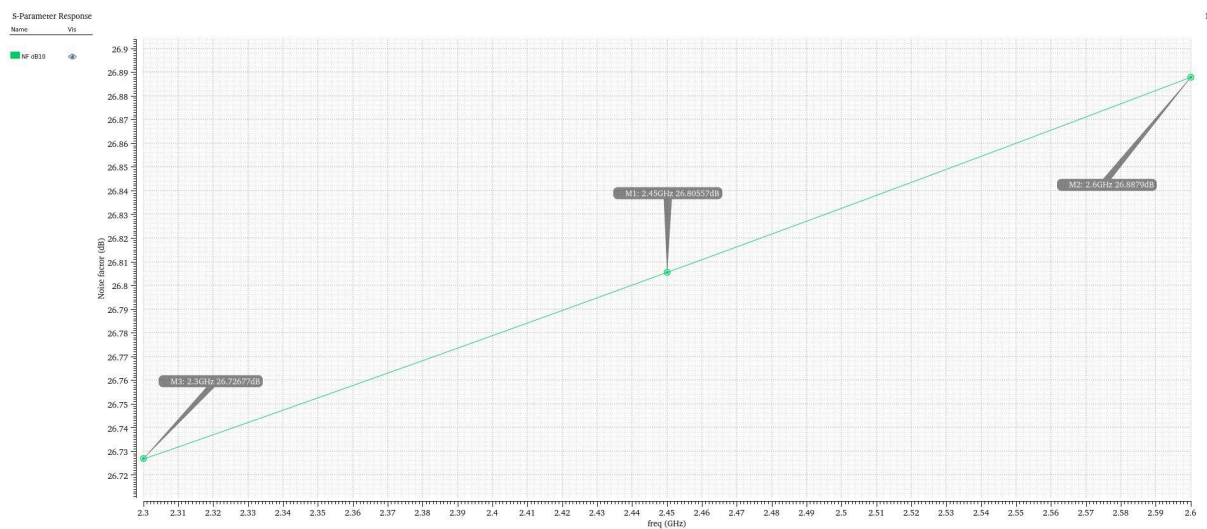


Figure 25: Noise for Standalone Baseband Amplifier 2.45 GHz

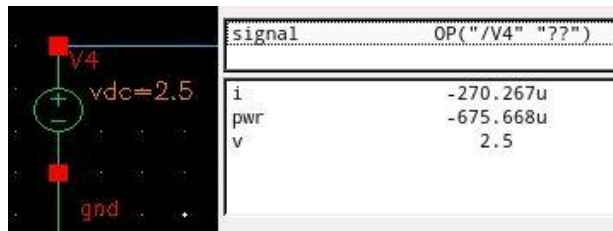


Figure 26: Power Consumption at 2.45 GHz

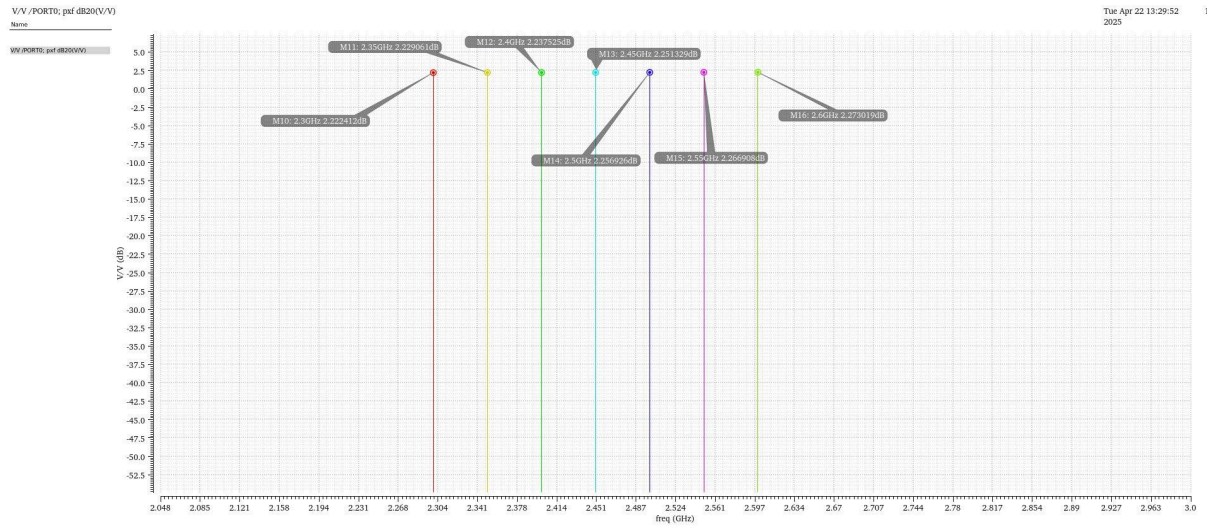


Figure 27: Conversion Gain for Swept LO and Fixed IF at 2.6 GHz

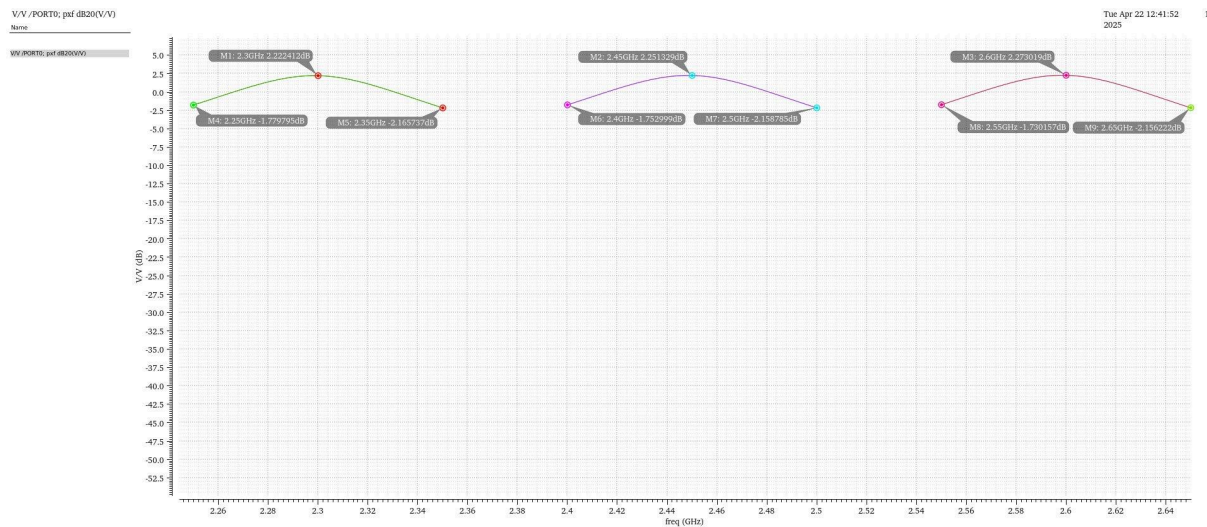


Figure 28: Conversion Gain for Swept IF and Fixed LO at 2.6 GHz

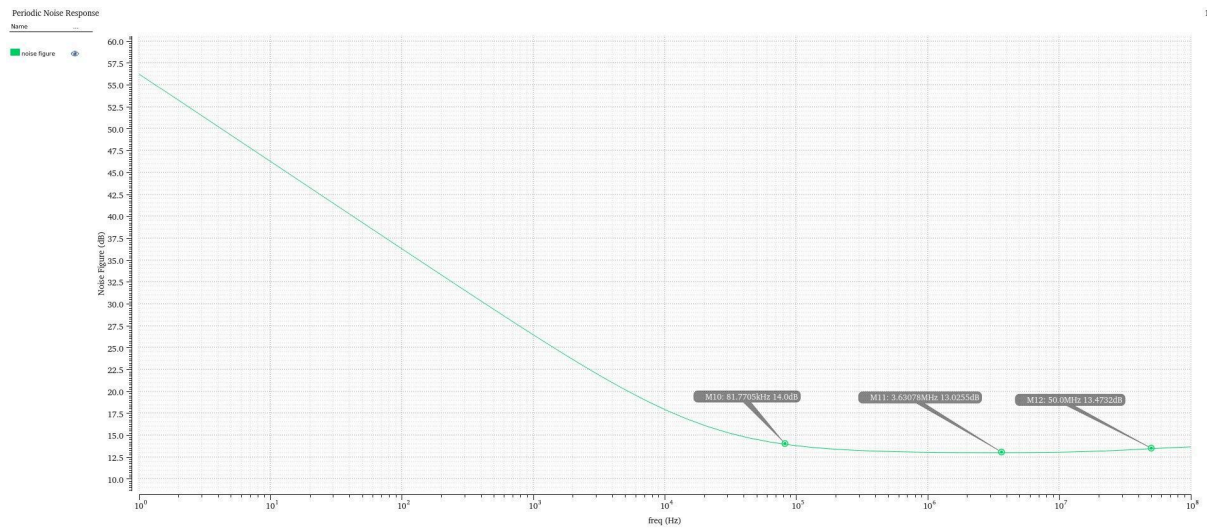


Figure 29: Noise Figure vs Baseband Frequency at 2.6 GHz (Meets requirement at baseband frequency. Flicker noise occurs between 0-781 kHz)

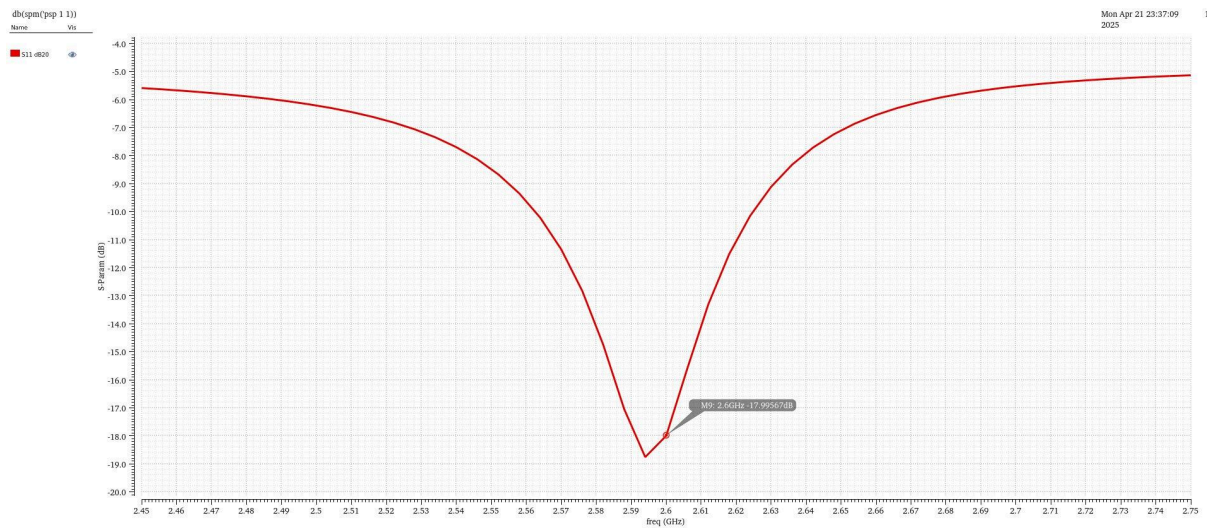


Figure 30: S11 Magnitude vs Frequency at 2.6 GHz

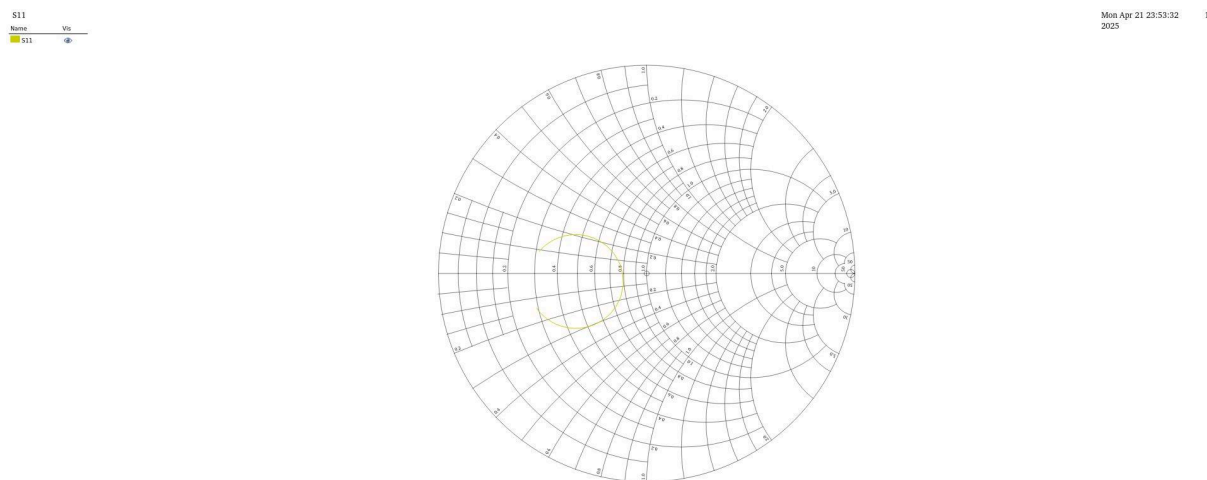


Figure 31: S11 Smith Chart at 2.6 GHz

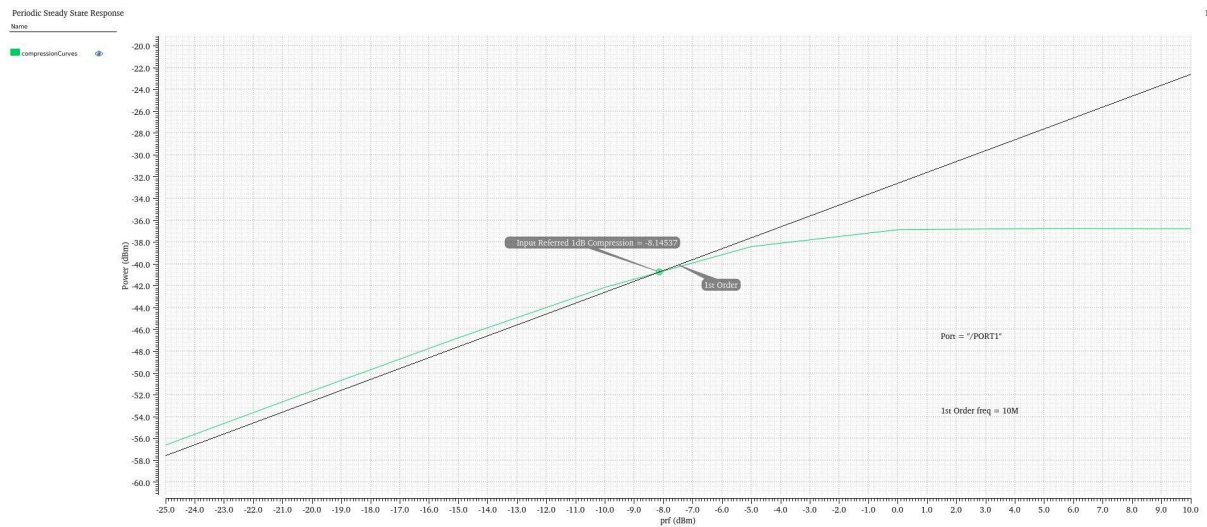


Figure 32: Input Referred P1dB Compression Point at 2.6 GHz

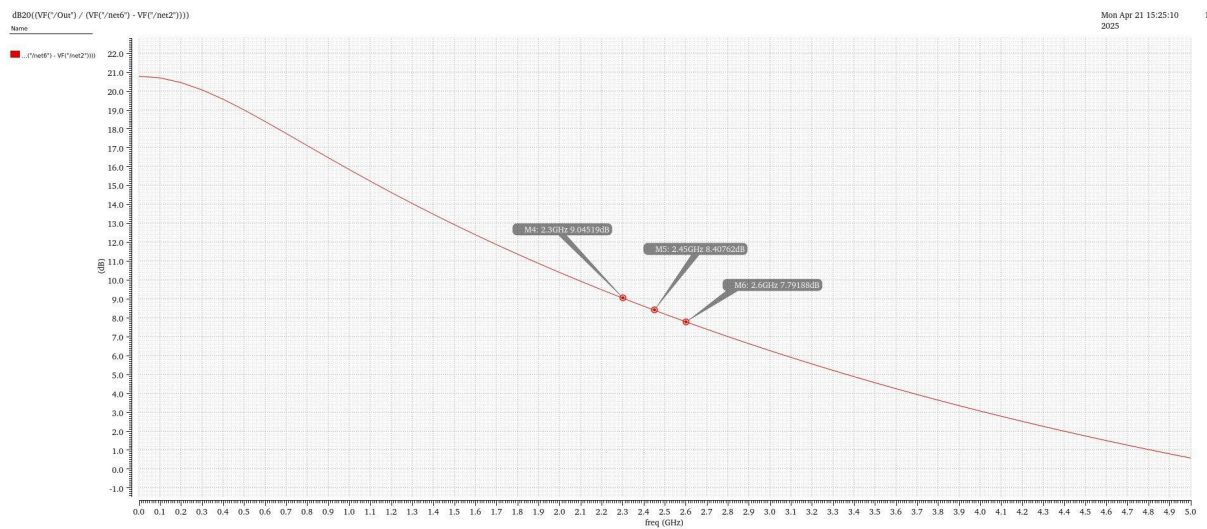


Figure 33: AC Gain for Standalone Baseband Amplifier at 2.6 GHz

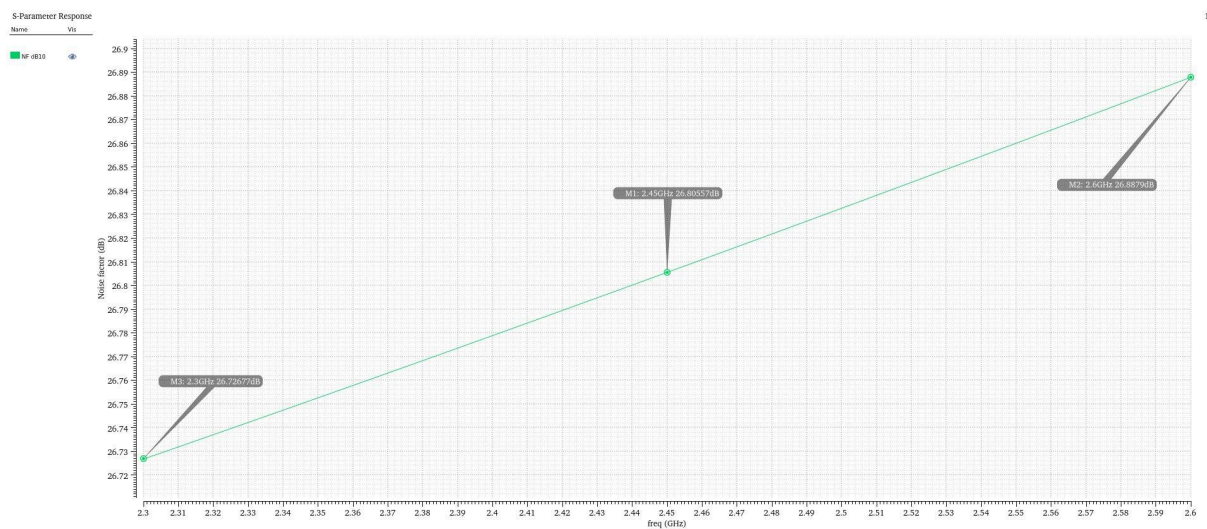


Figure 34: Noise for Standalone Baseband Amplifier 2.6 GHz

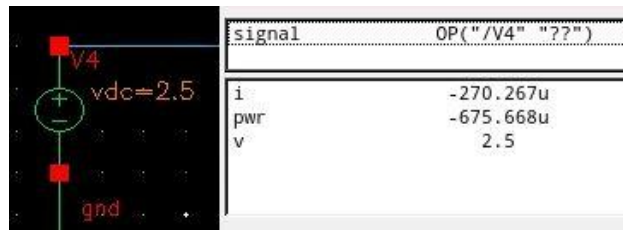


Figure 35: Power Consumption at 2.6 GHz

## References:

- 1) ECE 511 Lecture 09 - Active Loads - Dr. Brian Floyd
- 2) ECE 712 Lecture 16-19 - Mixers - Dr. Brian Floyd .
- 3) [https://web.ece.ucsb.edu/Faculty/rodwell/Courses/ece218c/tutorials\\_etc/Mixers.pdf](https://web.ece.ucsb.edu/Faculty/rodwell/Courses/ece218c/tutorials_etc/Mixers.pdf)
- 4) <https://onlinelibrary.wiley.com/doi/pdf/10.1002/9781118999745.app1>